

1. Administrative & Study Identification

Full Official Title A Two-Stage Double-Blind, Randomized, Placebo-Controlled Study to Assess the Efficacy, Safety and Pharmacokinetics of Alpelisib in Pediatric and Adult Patients with Lymphatic Malformations Associated with a PIK3CA Mutation
Study Title Alpelisib in Pediatric and Adult Patients With Lymphatic Malformations Associated With a PIK3CA Mutation. **Short Title/Acronym** EPIK-L1 **Protocol Number** CBYL719P12201 **NCT Number** NCT05948943 **Posting ID** 160494253867988138 **Sponsor / Company** Novartis **Investigational Product** Alpelisib (film-coated tablets, drinkable suspension, granules, oral suspension) | **Trade Names:** Viojoy, Piquay | **ATC Code:** L01EM03 **Phase of Development** PHASE2, PHASE3 **Trial Status** RECRUITING **Medical Monitor** Novartis Clinical Operations Lead / Study Contact

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To demonstrate the efficacy of alpelisib based on the radiological response rate (defined as a $\geq 20\%$ reduction from baseline in the sum of target lesion volumes) at Week 24 of Stage 2. **Secondary Objectives:** To evaluate Patient Global Impression of Severity (PGI-S) score (≥ 1 -point improvement compared to baseline), safety, tolerability, duration of response, pharmacokinetics (PK), and the frequency of LyM-related symptoms/complications over time.

2.2 Background & Rationale Lymphatic malformations (LyM) are rare, serious conditions characterized by non-malignant masses of fluid-filled cysts resulting from the abnormal development of the lymphatic system. They impair quality of life through pain, hemorrhage, infections, and anatomical distortion. Most LyM cases (75%-80%) are driven by PIK3CA mutations (which encode PI3K α). Currently, no approved pharmacologic therapies exist. This study evaluates alpelisib, a PI3K α -specific inhibitor, to determine if it can safely reduce lesion size and symptom severity in pediatric and adult patients.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional (Seamless Phase 2/3 study with 2 stages) **Control Type:** Placebo-controlled (during Stage 2 confirmatory phase) **Blinding:** Double-blind (Stage 2) / Open-label (Stage 1 and extensions) **Randomization:** Stage 1: 1:1 ratio (dose selection); Stage 2: 2:1 ratio (Alpelisib to Placebo)

3.2 Planned Enrollment & Duration Targeted Enrollment: 232 evaluable patients **Number of Sites:** Global, multi-center (Europe, US, RoW) **Estimated Study Start:** 17-Nov-2023 **Estimated Primary Completion:** 31-May-2028

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male or female patients 0 years of age and older with a physician-confirmed and documented diagnosis of symptomatic lymphatic malformations (LyM) associated with a somatic PIK3CA mutation. 2. Participant has at least 1 measurable LyM lesion confirmed by a Blinded Independent Review Committee (BIRC) assessment prior to randomization. 3. Participant is not considered as a candidate for or is not willing to receive non-drug therapies including but not limited to sclerotherapy, embolization, and surgery until the completion of Week 24 in Stage 1 and 2. 4. Participants must be able to ingest study drug (either in tablet form or as a drinkable suspension, granules, or as an oral suspension).

4.2 Exclusion Criteria 1. Physician-confirmed diagnosis of PIK3CA-Related Overgrowth Spectrum (PROS), central conducting lymphatic anomaly, generalized lymphatic anomaly, Gorham-Stout disease, or Kaposiform lymphangiomatosis. 2. Established diagnosis of type I diabetes mellitus or uncontrolled type II diabetes mellitus, or known history of Stevens-Johnson syndrome, erythema multiforme, or toxic epidermal necrolysis. 3. Prior treatment with alpelisib or other PI3K inhibitors lasting ≥ 2 weeks at the time of informed consent.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Alpelisib dosing cohorts by age group (e.g., adults starting at 125 mg or 250 mg; pediatrics starting at 50 mg or 125 mg) administered daily. **Route of Administration:** Oral (film-coated tablets, drinkable suspension, oral suspension, or granules). Drug administration via feeding tube is permitted. **Duration of Treatment:** 24-week core/confirmatory phase, followed by an open-label extension phase. (Estimated individual participation lasts approx. 3 years).

5.2 Concomitant & Prohibited Therapy Permitted: Standard concomitant medications for symptom management. **Prohibited:** Other PI3K inhibitors; participants must be ineligible for or unwilling to undergo non-drug therapies (sclerotherapy, embolization, surgery) before week 24 of either study stage.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -28 to 0 Informed Consent, Genetic Confirmation, Baseline MRI, Blood Draws
Baseline	Day 0	Randomization, First Dose, Questionnaires	Interim
Weeks 4 to 12	Safety Labs, PK Blood Draws, ECGs, Diary Review	End of Core Study	Week 24 Primary Efficacy Assessment (MRI), PGI-S score, Entry into Open-Label Extension

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Radiological response rate, defined by achieving at least a 20% reduction in the sum of target lesion volumes (1 to 3 lesions) at Week 24 of Stage 2 (adult and pediatric 6-17 years of age participants). **Measurement Method:** Magnetic Resonance Imaging (MRI) assessed centrally by a Blinded Independent Review Committee (BIRC).

7.2 Secondary & Exploratory Endpoints 1. Change from baseline in Patient Global Impression of Severity (PGI-S) score (≥ 1 -point improvement). 2. Change from baseline in patient global impression of change (PGI-C) scale. 3. Percentage of participants with a radiological response at Week 24 of Stage 2 (pediatric participants 0-5 years of age).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine collection via patient diaries, interviews, and lab assessments (particularly monitoring for PI3K-related toxicities such as hyperglycemia and rash). **Serious Adverse Events (SAEs):** Standard reporting timeline to the sponsor within 24 hours. **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee to conduct periodic safety and efficacy reviews to assess the risk-benefit profile.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for efficacy endpoints; Safety Set for all participants receiving at least one dose of the investigational product. **Sample Size Justification:** Total N = 232 (Targeted Enrollment) properly powered to detect a statistically significant radiological response compared to placebo. **Handling of Missing Data:** Missing data handled via multiple imputation or non-responder imputation models based on the endpoint structure.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
Monitoring Plan: Combination of onsite and remote monitoring visits to ensure protocol and regulatory compliance. **Product Information:** [EMA EPAR Product Information](#) **Audit Trail:** Validated 21 CFR Part 11 compliant electronic data capture (EDC) system with eCRF locking and query resolution workflows.

11. Study Identifiers & Governance

Primary ID: {{CBYL719P12201}} **Posting ID:** {{160494253867988138}}
Registry Number: {{NCT05948943}} **Sponsor/Company:** {{Novartis}}
Study Title: {{Alpelisib in Pediatric and Adult Patients With Lymphatic Malformations Associated With a PIK3CA Mutation.}}
Official Regulatory Title: {{A Two-Stage Double-Blind, Randomized, Placebo-Controlled Study to Assess the Efficacy, Safety and Pharmacokinetics of Alpelisib in Pediatric and Adult Patients with Lymphatic Malformations Associated with a PIK3CA Mutation}} **Trial Status:** {{RECRUITING}}

12. Clinical Framework (Lymphatic Malformation Specific)

Disease Area / Condition: {{Lymphatic Malformations}} **Preferred Name:** {{Lymphatic Abnormalities}} **Preferred UMLS Name:** {{PIK3CA-related overgrowth syndrome}} **Semantic Type:** {{Anatomical Abnormality}} **Definition:** {{Congenital or acquired structural abnormalities of the lymphatic system (LYMPHOID TISSUE) including the lymph vessels.}} **MeSH Code:** {{D044148}} **SNOMED ID:** {{234095009}} **Diagnosis Threshold:** {{Confirmed diagnosis of LyM with documented somatic PIK3CA mutation}} **Medical Methodology:** {{Targeted PI3K α -specific inhibition (Alpelisib)}}
Optimization Target: {{ $\geq 20\%$ reduction in target lesion volume}}

13. Study Procedures & Methodology

Management Pathway: {{Targeted oncology and rare disease management pathway for LyM}} **Education Protocol (HCP):** {{PI3K α inhibitor safety, dosing, and AE management training (e.g., hyperglycemia)}} **Education Protocol (Patient):** {{Study drug administration (tablet/suspension/granules) and AE symptom tracking/diaries}} **Quality Audit Mechanism:** {{Blinded Independent Review Committee (BIRC) assessment of MRI and clinical response}}

14. Observation & Follow-Up Schedule

Total Duration: {{24-week core phase followed by an open-label extension (approx. 3 years participation)}} **Onsite Visit Cadence:** {{Baseline, Week 12, Week 24, and subsequent extension visits}} **Remote Monitoring Frequency:** {{Continuous AE tracking / symptom diaries}} **Checklist Review Interval:** {{Questionnaires and diaries collected at scheduled intervals (e.g., Week 24)}}

15. Sign-off & Approval

Role	Name	Signature	Date		----	----	----	----
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					